Analytical Chemistry Cumulative Examination
October 14, 2011

1. Electrospray ionization mass spectrometry is an exceptionally useful technique for analysis of high mass compounds. Please answer the following questions about this technique.

   (a) Prior to introduction of this method, quadrupole mass analyzers were generally limited to analysis of molecules with molecular weights of 2000 Daltons or lower. ESI-MS has greatly changed this situation, particularly for proteins. What are the current mass limits for protein analysis by ESI-MS? Explain. (10 points)

   (b) Data processing of electrospray mass spectra can have a number of complications. Please explain what they are and what methods (both computational and experimental) can be used to overcome these problems. (15 points)

   (c) One model for the electrospray ionization process is the Iribarne and Thomson ion extraction model. Explain with a clearly labeled diagram the basic idea of their mechanism. (25 points)

2. MALDI mass spectrometry is another exceptionally useful method. In July,2011 Trimpin and coworkers published an interesting paper in Analytical Chemistry entitled "Extending the Laserspray Ionization Concept to Produce Highly Charged Ions at High Vacuum on a Time-of-Flight Mass Analyzer". What is laserspray ionization as they defined it? What are the advantages of the technique? A new matrix was introduced in this paper. What is that matrix? Evaluate the utility of the new method. (50 points)